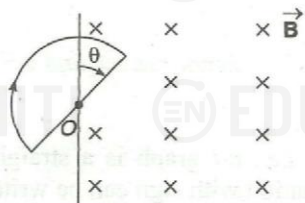
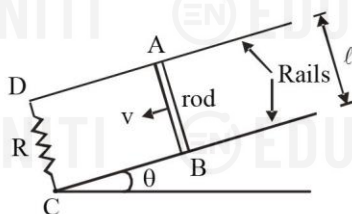


1. A wire loop enclosing a semicircle of radius R is located on the boundary of uniform magnetic field B . At the moment $t = 0$, the loop is set into rotation with a constant angular acceleration α about an axis O coinciding with a line of vector $\vec{B}$ on the boundary. Find the emf induced in the loop as a function of time. Draw the approximate plot of this function. The arrow in the figure shows the emf direction taken to be positive.



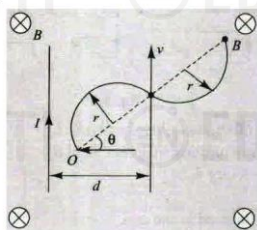
2. Two rails AD and BC are fixed parallel to each other separated by distance $\ell = AB = CD$ and inclined at angle θ with horizontal as shown in the figure.



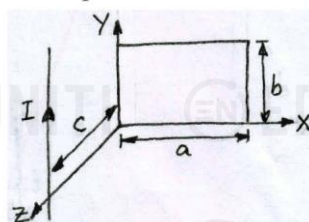
The lower ends of the rails C and D are connected by a resistor R . A copper rod of mass m , slides down the rail under the influence of gravity. A uniform vertically downward magnetic field B exists throughout the region. Show that the rod acquires a steady-state velocity

$$v = \frac{mgR \sin \theta}{B^2 \ell^2 \cos^2 \theta}$$

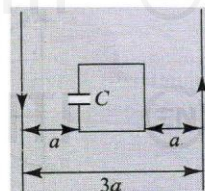
3. An infinite wire carries a current I . An S-shaped conducting rod of two semicircles each of radius r is placed at an angle θ to the wire. The center of the conductor is at a distance d from the wire. If the rod translates parallel to the wire with a velocity v as shown in fig., calculate the emf induced across the ends OB of the rod.



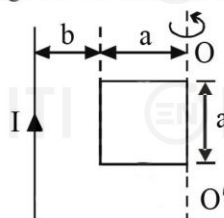
4. In fig., a long thin wire with a varying current $I = I_0 \cos \omega t$ lies at a distance c from one edge of a rectangular wire loop of length a and width b lying in the x - y plane. What emf is induced in the loop?



5. Two infinitely long parallel wires carrying currents $I = I_0 \sin \omega t$ in opposite directions are placed at a distance $3a$ apart. A square loop of side a of negligible resistance with a capacitor of capacitance C is placed in the plane of wires as shown in fig. Find the maximum current in the square loop. Also find the instantaneous charge on the upper plate of the capacitor and sketch the graph showing its variation as a function of time for one complete cycle taking anticlockwise direction of the current in the loop as positive.



6. A square loop of side a and a straight infinite conductor carrying current I are placed in the same plane with two sides of the square parallel to the conductor. The inductance and resistance of the loop are equal to L and R respectively. The frame is turned through 180° about the axis OO' and stopped. Find the electric charge that flows in the square loop.

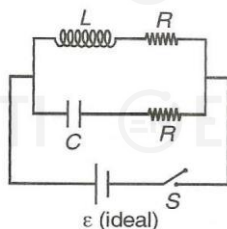


7. A small bar magnet of dipole moment $\vec{M}$ is pointing and moving with a constant velocity v along x -axis. A small closed circular conducting loop of radius a and negligible self-inductance lies in the y - z plane with its centre at $x = 0$, and its axis coinciding with the x -axis. Find the force opposing the motion of the magnet, if the resistance of the loop is R . Assume that the distance x of the magnet from the centre of the loop is much greater than a .

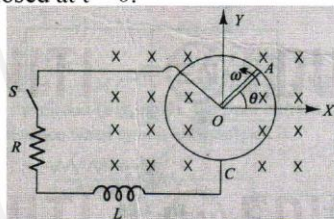
8. Consider a LR series circuit in which the switch is closed at $t=0$. Let P_L be the rate at which energy increases in the inductor and P_R be the rate at which energy is dissipated as heat in the resistor.

- Express the ratio P_L / P_R as a function of time and the time constant τ of circuit.
- What is the time at which $P_L = P_R$?

9. A circuit is shown in the figure. The switch is closed at $t = 0$. Find the condition for which current drawn from the battery does not vary with time.



10. In the arrangement shown in fig., a metal rod OA of mass m and length r is made to rotate with a constant angular speed ω in a vertical plane about a horizontal axis at the end O . The free end A slides without friction on a fixed conducting circular ring. A uniform magnetic field B exists perpendicular to the plane of rotation. Neglect the resistance of the ring and the rod. The switch is closed at $t = 0$.



- What is the induced emf across the terminals of the switch before it is closed.
 - After the switch S is closed obtain an expression for the current as a function of time
 - When steady state is reached, obtain the time dependence of the torque required to maintain the constant angular speed.
- Consider the rod OA to be along the positive X -axis at $t = 0$